



give me viva questions and ans for ai practical according to sppu comp engg 2019 pattern

main topics -

dfs,bfs

a star 8 puzzle algo

greedy algo - prims, kruskal,dijkstra, selection sort

csp - n queens, graph coloring

basic rule based chatbot

Here is a comprehensive set of AI Practical Viva Questions & Answers tailored for SPPU Computer Engineering 2019 Pattern:

☒ DFS & BFS

Q1. What is the difference between DFS and BFS?

- BFS explores all neighbors at the current depth before going deeper; it uses a **Queue** ^[1]
- DFS explores as deep as possible before backtracking; it uses a **Stack or Recursion** ^[1]
- BFS guarantees the **shortest path** in an unweighted graph; DFS does not ^[2]

Q2. What data structure is used in BFS and DFS?

- BFS → **Queue (FIFO)**
- DFS → **Stack (LIFO)** or call stack via recursion ^[1]

Q3. What are the time and space complexities of BFS and DFS?

- Both: Time = $O(V + E)$, Space = $O(V)$ where V = vertices, E = edges ^[1]

Q4. Is BFS or DFS better for finding the shortest path?

- BFS is better since it explores level by level, guaranteeing the minimum number of steps ^[2]

Q5. What are real-world applications of BFS and DFS?

- BFS: Social network friend suggestions, GPS navigation, web crawlers
- DFS: Topological sorting, cycle detection, maze solving ^[1]

Q6. Can DFS be used on a disconnected graph?

- Yes, by calling DFS for each unvisited vertex to cover all components ^[1]

★ A* Algorithm & 8-Puzzle

Q7. What is the A algorithm?*

- A* is an informed search algorithm that finds the shortest path using the evaluation function: $f(n) = g(n) + h(n)$ where $g(n)$ = actual cost from start and $h(n)$ = heuristic estimate to goal [2]

Q8. What heuristic is used in the 8-puzzle problem?

- Manhattan Distance – sum of horizontal + vertical distances of each tile from its goal position
- Misplaced Tiles – count of tiles not in their goal position [2]

Q9. What is the 8-puzzle problem?

- A 3×3 grid with 8 numbered tiles and 1 blank; the goal is to reach a target configuration by sliding tiles [2]

Q10. Why is A preferred over BFS for the 8-puzzle?*

- BFS explores blindly in all directions; A* uses a heuristic to guide the search, making it significantly faster and more memory-efficient [2]

Q11. What does it mean for A to be admissible?*

- A heuristic is admissible if it never overestimates the actual cost; this guarantees A* finds the optimal solution [2]

Q12. What is the difference between A and Greedy Best-First Search?*

- Greedy only uses $h(n)$ and is not guaranteed optimal; A* uses both $g(n) + h(n)$ ensuring optimal solution [2]

☒ Greedy Algorithms

Selection Sort

Q13. How does Selection Sort work?

- Finds the minimum element from the unsorted part and swaps it with the first element of that part, repeating until sorted [3]

Q14. What is the time complexity of Selection Sort?

- $O(n^2)$ in all cases — it always scans the entire remaining list for the minimum [3]

Q15. Is Selection Sort stable?

- No, it is not a stable sort [4]

Q16. Why is Selection Sort considered a greedy algorithm?

- At each step it greedily picks the locally minimum element, building the sorted array one element at a time ^[1]

Prim's Algorithm

Q17. What does Prim's algorithm do?

- Builds a Minimum Spanning Tree (MST) by starting from any node and repeatedly adding the minimum weight edge connecting the tree to an unvisited vertex ^[4]

Q18. What data structure is used in Prim's algorithm?

- A priority queue / Min-Heap and an `mstSet[]` array to track included vertices ^[4]

Q19. What is the time complexity of Prim's algorithm?

- $O(E \log V)$ with a priority queue; $O(V^2)$ with adjacency matrix ^[1]

Q20. When is Prim's preferred over Kruskal's?

- Prim's is better for dense graphs (many edges) ^[1]

Kruskal's Algorithm

Q21. How does Kruskal's algorithm work?

1. Sort all edges by weight ^[1]
2. Pick the smallest edge that doesn't form a cycle (using Disjoint Set Union / Union-Find) ^[1]
3. Repeat until $V-1$ edges are included ^[1]

Q22. What is the time complexity of Kruskal's algorithm?

- $O(E \log E)$ due to edge sorting ^[1]

Q23. How does Kruskal's avoid cycles?

- Using the Union-Find (DSU) data structure; if both endpoints of an edge belong to the same set, adding it would form a cycle ^[1]

Q24. Can a graph have more than one MST?

- Yes, if multiple spanning trees have the same total weight ^[1]

Dijkstra's Algorithm

Q25. What is Dijkstra's algorithm used for?

- Finding the Single Source Shortest Path from a source vertex to all other vertices in a weighted graph ^[5]

Q26. What is the time complexity of Dijkstra's algorithm?

- $O(E \log V)$ with a priority queue; $O(V^2)$ with adjacency matrix^[1]

Q27. Does Dijkstra's work with negative weights?

- No, Dijkstra's fails with negative edge weights; use **Bellman-Ford** instead^[1]

Q28. What is the difference between Dijkstra's and Floyd-Warshall?

- Dijkstra's: single-source shortest path, $O(E \log V)$
- Floyd-Warshall: all-pairs shortest path, $O(V^3)$, handles negative weights^[1]

☒ CSP – N-Queens & Graph Coloring

Q29. What is a Constraint Satisfaction Problem (CSP)?

- A problem defined by a set of variables, each with a domain of values, and a set of constraints that the solution must satisfy^[5]

Q30. Explain the N-Queens problem.

- Place N queens on an $N \times N$ chessboard such that no two queens attack each other (same row, column, or diagonal)^[5]

Q31. What technique is used to solve N-Queens?

- **Backtracking with Branch and Bound** — if placing a queen leads to a conflict, backtrack and try the next position^[5]

Q32. What is the time complexity of the N-Queens solution?

- $O(N!)$ in the worst case; Branch and Bound significantly prunes the search space^[5]

Q33. What is Graph Coloring?

- Assigning colors to graph vertices such that no two adjacent vertices share the same color; the minimum number of colors needed is the **chromatic number**^[5]

Q34. What algorithm solves Graph Coloring in AI?

- **Backtracking** — assign a color to a vertex, check constraints (no adjacent vertex has the same color), and backtrack if a conflict arises^[5]

Q35. What is the difference between N-Queens and Graph Coloring as CSPs?

- N-Queens constraints involve rows, columns, and diagonals; Graph Coloring constraints involve edge adjacency — but both use backtracking as the core strategy^[5]

☒ Rule-Based Chatbot

Q36. What is a rule-based chatbot?

- A chatbot that responds to user inputs using **predefined rules**, typically pattern matching using if-else conditions or regular expressions — no machine learning involved

Q37. What are the limitations of a rule-based chatbot?

- Cannot handle unseen inputs or typos, no contextual understanding, doesn't learn from conversations, difficult to scale with complex conversations

Q38. What is the difference between a rule-based and AI-based chatbot?

- Rule-based: uses hardcoded patterns/responses, deterministic
- AI-based: uses NLP/ML models (e.g., BERT, GPT), can generalize to new inputs

Q39. What Python library is commonly used to build a rule-based chatbot?

- NLTK (Natural Language Toolkit) with Chat class and pattern-response pairs

Q40. What is pattern matching in a chatbot?

- Comparing user input against a list of regex patterns; when a match is found, the corresponding response is returned

☒ General AI Viva

Q41. What is the difference between informed and uninformed search?

- Uninformed (blind): BFS, DFS — no knowledge about goal location
- Informed (heuristic): A*, Greedy BFS — use domain knowledge via heuristic function ^[2]

Q42. What is a heuristic function?

- An estimate of the cost from the current node to the goal; a good heuristic is admissible (never overestimates) and consistent ^[2]

Q43. What is backtracking?

- A strategy where the algorithm tries all possible options and undoes (backtracks) the last choice when it reaches a dead end ^[5]

Q44. What is the state space in AI search?

- The set of all possible configurations/states a problem can be in; search algorithms traverse this space to find the goal state ^[2]

Q45. What is the difference between a spanning tree and MST?

- A spanning tree connects all vertices with V-1 edges; an MST specifically has the minimum total edge weight ^[1]

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1. <https://www.scribd.com/document/835487583/Viva-Questions-and-Answers-on-Design-and-Analysis-of-Algorithms>

2. <https://www.geeksforgeeks.org/8-puzzle-problem-in-ai/>

3. <https://www.scribd.com/document/895064489/Viva-Questions-Answers>

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